

VU Minor Applied Econometrics
Bayesian Econometrics for Business & Economics
(Bayesian statistics & simulation methods)
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Lecture 4:

- Exercise 4: Bernoulli distributed data:
posterior distribution of θ in case of informative, conjugate prior.
- Simulation method: random walk Metropolis(-Hastings) method:
 - Application to posterior density kernel $P(\theta) = p(\theta)p(y|\theta)$ in Autoregressive Conditional Heteroskedasticity (ARCH) model.
(Slides 8-15 and 16-20 are slides 20-27 and 43-47 from lecture 3.)
 - Application to posterior density kernel $P(\theta) = p(\theta)p(y|\theta)$ in Poisson regression model (with 2-dimensional $\theta = (\beta_0, \beta_1)'$).
 - Application to uniform target density $P(\theta)$. Purely for illustration!
(Slides 33-41 are slides 34-42 from lecture 3.)

Exercise 4: Bernoulli model under informative, conjugate prior

Model: Bernoulli distribution:

- $y_i = 1$ with probability θ
- $y_i = 0$ with probability $1 - \theta$

Likelihood:

$$\begin{aligned} p(y|\theta) &= p(y_1, \dots, y_n|\theta) = \prod_{i=1}^n p(y_i|\theta) = \theta^{\sum_{i=1}^n y_i} (1 - \theta)^{\sum_{i=1}^n (1 - y_i)} \\ &= \theta^{n_1} (1 - \theta)^{n_0} \end{aligned}$$

with $n_1 \equiv \sum_{i=1}^n y_i$ the number of ones in the sample, and
 with $n_0 \equiv \sum_{i=1}^n (1 - y_i)$ the number of zeros in the sample.

Informative prior: $Beta(\tilde{n}_1 + 1, \tilde{n}_0 + 1)$ distribution:

$$p(\theta) = \begin{cases} \frac{\Gamma(\tilde{n}_1 + \tilde{n}_0 + 2)}{\Gamma(\tilde{n}_1 + 1)\Gamma(\tilde{n}_0 + 1)} \theta^{\tilde{n}_1} (1 - \theta)^{\tilde{n}_0} & \text{if } 0 \leq \theta \leq 1, \\ 0 & \text{else.} \end{cases}$$

This is a **conjugate** prior: a prior such that prior and posterior are in the same probability distribution family (that is, a prior that has the specification like a posterior based on an older dataset).

For example, $\tilde{n}_1 = 2$ and $\tilde{n}_0 = 8$ would have the same effect as adding 10 artificial observations (with two “successes” and eight “failures”) to our actual dataset.

(a) Suppose that we specify this $Beta(\tilde{n}_1 + 1, \tilde{n}_0 + 1)$ prior distribution, and that we have a dataset of n observations with n_1 “successes” and n_0 “failures”. Use Bayes’ rule to derive a kernel of posterior density $p(\theta|y)$.

Answer: Prior:

$$p(\theta) \propto \begin{cases} \theta^{\tilde{n}_1} (1 - \theta)^{\tilde{n}_0} & \text{if } 0 \leq \theta \leq 1, \\ 0 & \text{else.} \end{cases}$$

Likelihood:

$$p(y|\theta) = \theta^{n_1} (1 - \theta)^{n_0}.$$

Posterior density kernel:

$$p(\theta|y) \propto p(\theta)p(y|\theta) \propto \begin{cases} \theta^{\tilde{n}_1 + n_1} (1 - \theta)^{\tilde{n}_0 + n_0} & \text{if } 0 \leq \theta \leq 1, \\ 0 & \text{else.} \end{cases}$$

Note: same posterior density kernel as in case where we have a uniform prior and where we observe $\tilde{n}_1 + n_1$ “successes” and $\tilde{n}_0 + n_0$ “failures”.

(b) What is the exact posterior density $p(\theta|y)$, including the scaling factor? What is the posterior mean $E(\theta|y)$? You can make use of Table 1a-1b.

Answer: Posterior density kernel:

$$p(\theta|y) \propto p(\theta)p(y|\theta) \propto \begin{cases} \theta^{\tilde{n}_1+n_1}(1-\theta)^{\tilde{n}_0+n_0} & \text{if } 0 \leq \theta \leq 1, \\ 0 & \text{else.} \end{cases}$$

This is a $\text{Beta}(\tilde{n}_1 + n_1 + 1, \tilde{n}_0 + n_0 + 1)$ distribution with density:

$$p(\theta|y) = \begin{cases} \frac{\Gamma(\tilde{n}_1+n_1+\tilde{n}_0+n_0+2)}{\Gamma(\tilde{n}_1+n_1+1)\Gamma(\tilde{n}_0+n_0+1)} \theta^{\tilde{n}_1+n_1}(1-\theta)^{\tilde{n}_0+n_0} & \text{if } 0 \leq \theta \leq 1, \\ 0 & \text{else.} \end{cases}$$

The posterior mean is $E(\theta|y) = \frac{\tilde{n}_1+n_1+1}{\tilde{n}_1+n_1+\tilde{n}_0+n_0+2}$.

Table 1a: Continuous distributions

distribution	probability density function	mean
Beta(a, b)	$p(x) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} x^{a-1} (1-x)^{b-1}$ for $0 \leq x \leq 1$	$\frac{a}{a+b}$
Exponential(b)	$p(x) = \frac{1}{b} \exp\left(-\frac{x}{b}\right)$ for $x \geq 0$	b
Gamma(a, b)	$p(x) = \frac{1}{\Gamma(a)b^a} x^{a-1} \exp\left(-\frac{x}{b}\right)$ for $x \geq 0$	$a \cdot b$
Normal $N(\mu, \sigma^2)$	$p(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$ for $-\infty < x < \infty$	μ
Student-t(μ, σ^2, DoF)	$p(x) = \frac{\Gamma(\frac{DoF+1}{2})}{\Gamma(\frac{DoF}{2})\sqrt{DoF}\pi} \frac{1}{\sigma} \left(1 + \frac{(x-\mu)^2}{DoF\sigma^2}\right)^{-\frac{DoF+1}{2}}$ for $-\infty < x < \infty$	μ

Overview of integration methods:

integration

⋮
analytical

⋮
numerical

⋮
deterministic
(possible for
 $\text{dim. } \theta \leq 3$)

⋮
simulation
(Monte Carlo)

⋮
direct
simulation

⋮
indirect
simulation

⋮
well-known
conditional
posteriors:

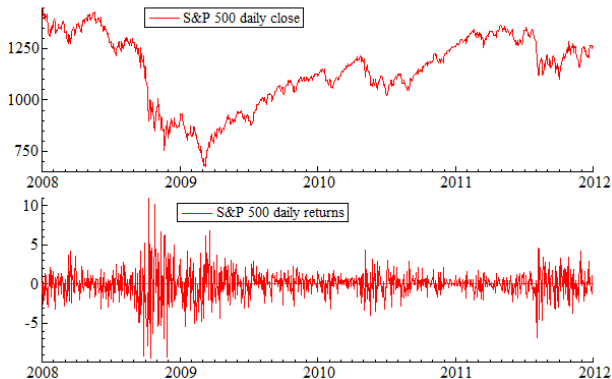
⋮
Gibbs sampling

⋮
unknown
conditional
posteriors:

⋮
Metropolis-Hastings:
random walk
Metropolis(-Hastings),
independence chain
Metropolis-Hastings.

Example: Autoregressive Conditional Heteroskedasticity (ARCH) model

Data: S&P 500 daily close p_t and log-returns $y_t = 100 \times \ln(\frac{p_t}{p_{t-1}})$:



Note: **volatility clustering**: consecutive periods with large variance and consecutive periods with small variance.

Simple case: ARCH(1) model with mean 0 and normal distribution:

$$y_t | I_{t-1} \sim N(0, \sigma_t^2)$$

with conditional variance

$$\sigma_t^2 = \text{var}(y_t | I_{t-1}) = \alpha_0 + \alpha_1 y_{t-1}^2$$

with information set $I_{t-1} = \{y_{t-1}, y_{t-2}, \dots\}$.

Two parameters:

- α_0 ($\alpha_0 > 0$): constant term in variance equation
- α_1 ($0 \leq \alpha_1 < 1$): effect of yesterday's squared return y_{t-1}^2 on today's return's variance $\text{var}(y_t | I_{t-1})$.

Note: the restrictions $\alpha_0 > 0$ and $\alpha_1 \geq 0$ ensure that $\alpha_0 + \alpha_1 y_{t-1}^2 > 0$.

Simple case: ARCH(1) model with mean 0 and normal distribution:

$$y_t | I_{t-1} \sim N(0, \sigma_t^2) \quad \sigma_t^2 = \text{var}(y_t | I_{t-1}) = \alpha_0 + \alpha_1 y_{t-1}^2.$$

The unconditional variance is:

$$\text{var}(y_t) = \frac{\alpha_0}{1 - \alpha_1}$$

(Derivation:

$$\begin{aligned} \text{var}(y_t) &= E(y_t^2) \\ &= E(E(y_t^2 | I_{t-1})) \\ &= E(\alpha_0 + \alpha_1 y_{t-1}^2) \\ &= \alpha_0 + \alpha_1 E(y_{t-1}^2) \\ &= \alpha_0 + \alpha_1 \text{var}(y_{t-1}) \\ &= \alpha_0 + \alpha_1 \text{var}(y_t), \end{aligned}$$

where

$$\text{var}(y_t) = \text{var}(y_{t-1})$$

holds because the ARCH(1) process is stationary for $0 \leq \alpha_1 < 1$.)

Variance targeting: estimate model so that (estimated) unconditional variance is equal to sample variance $s^2 = \frac{1}{n-1} \sum_{t=1}^n (y_t - \bar{y})^2$.

Here in ARCH(1) model:

$$\frac{\alpha_0}{1 - \alpha_1} = s^2$$

$$\alpha_0 = s^2(1 - \alpha_1)$$

ARCH(1) model becomes:

$$y_t \sim N(0, \sigma_t^2) \quad \sigma_t^2 = \text{var}(y_t | I_{t-1}) = s^2(1 - \alpha_1) + \alpha_1 y_{t-1}^2$$

with only one parameter α_1 (good for *illustrative example* of model with 'non standard' posterior distribution!)

Conditional density of y_t given $y_{t-1}, y_{t-2}, \dots$:

$$\begin{aligned}
 p(y_t | y_{t-1}, \alpha_1) &= (2\pi\sigma_t^2)^{-1/2} \exp\left(-\frac{y_t^2}{2\sigma_t^2}\right) \\
 &= (2\pi[\alpha_0 + \alpha_1 y_{t-1}^2])^{-1/2} \exp\left(-\frac{y_t^2}{2[\alpha_0 + \alpha_1 y_{t-1}^2]}\right) \\
 &= (2\pi[s^2(1 - \alpha_1) + \alpha_1 y_{t-1}^2])^{-1/2} \exp\left(-\frac{y_t^2}{2[s^2(1 - \alpha_1) + \alpha_1 y_{t-1}^2]}\right)
 \end{aligned}$$

for

- any (G)ARCH model with normal density,
- ARCH(1) model with normal density, and
- ARCH(1) model with normal density with variance targeting,

respectively.

Likelihood (conditional on 'fixed' first observation y_1):

$$\begin{aligned}
 p(y_2, \dots, y_n | \alpha_1) &= \prod_{t=2}^n p(y_t | y_{t-1}, y_{t-2}, \dots, \alpha_1) \\
 &= \prod_{t=2}^n p(y_t | y_{t-1}, \alpha_1) \\
 &= \prod_{t=2}^n \left\{ (2\pi[s^2(1 - \alpha_1) + \alpha_1 y_{t-1}^2])^{-1/2} \times \right. \\
 &\quad \left. \exp\left(-\frac{y_t^2}{2[s^2(1 - \alpha_1) + \alpha_1 y_{t-1}^2]}\right) \right\}
 \end{aligned}$$

Prior: suppose we specify (non-informative) uniform prior on $[0,1]$ for α_1 :

$$p(\alpha_1) = \begin{cases} 1 & \text{if } 0 \leq \alpha_1 < 1, \\ 0 & \text{else.} \end{cases}$$

Posterior:

$$\begin{aligned}
 p(\alpha_1|y) &\propto p(y|\alpha_1)p(\alpha_1) \\
 &\propto \prod_{t=2}^n \left\{ [s^2(1 - \alpha_1) + \alpha_1 y_{t-1}^2]^{-1/2} \times \right. \\
 &\quad \left. \exp \left(-\frac{y_t^2}{2[s^2(1 - \alpha_1) + \alpha_1 y_{t-1}^2]} \right) \right\}
 \end{aligned}$$

if $0 \leq \alpha_1 < 1$; 0 else.

Note: this is **not** a well-known posterior distribution.

⇒ Use simulation method, for example **Metropolis-Hastings** method.

If we would have two parameters α_0 and α_1 , then the *conditional* posterior distributions would also **not** be well-known distributions.

⇒ Gibbs sampling **not** possible for Bayesian analysis of (Generalized) Autoregressive Conditional Heteroskedasticity ((G)ARCH) models.

Application to posterior density kernel in ARCH(1) model

Target density kernel $P(\alpha_1) = p(\alpha_1)p(y|\alpha_1)$ with logarithm

$$\ln P(\alpha_1) = \ln p(\alpha_1) + \ln p(y|\alpha_1)$$

with log-prior:

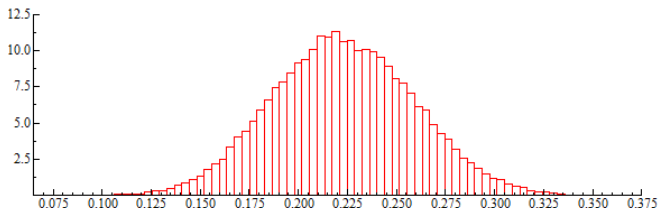
$$\ln p(\alpha_1) = \begin{cases} \ln(1) = 0 & \text{if } 0 \leq \alpha_1 \leq 1, \\ \ln(0) = -\infty & \text{else.} \end{cases}$$

and loglikelihood:

$$\begin{aligned} \ln p(y|\alpha_1) &= \ln \left(\prod_{t=2}^n p(y_t|y_{t-1}, \alpha_1) \right) = \sum_{t=2}^n \ln(p(y_t|y_{t-1}, \alpha_1)) = \\ &= \sum_{t=2}^n \left\{ -\frac{1}{2} \ln(2\pi[s^2(1 - \alpha_1) + \alpha_1 y_{t-1}^2]) - \frac{y_t^2}{2[s^2(1 - \alpha_1) + \alpha_1 y_{t-1}^2]} \right\} \end{aligned}$$

In our ARCH(1) model:

- Initial value $\theta_0 = \text{ML estimator}$.
- Candidate distribution: $\tilde{\theta} \sim N(\theta_{i-1}, 0.03^2)$
(normal distribution with small standard deviation 0.03).
- $n_{draws} = 100100$ draws (with burn-in of 1000 draws).

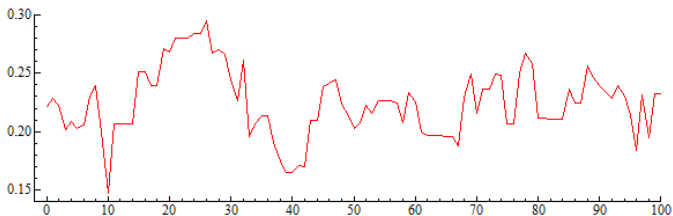


Posterior mean (stdev): 0.223 (0.036).

Maximum likelihood estimator (standard error): 0.221 (0.037)

Evaluation of quality of candidate distribution $\tilde{\theta} \sim N(\theta_{i-1}, 0.03^2)$:

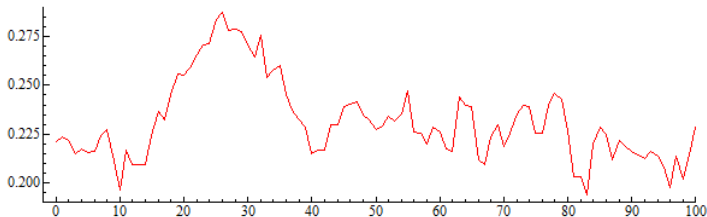
- **'trace plot'** of (accepted) draws $\theta_0, \theta_1, \theta_2, \dots$:



- **acceptance percentage:** what percentage of the candidate draws is accepted?
Here: acceptance percentage = 75.4% (rather high).
- **(first order) serial correlation in sequence of (accepted) draws.**
Here: $\text{corr}(\alpha_{1,i}, \alpha_{1,i-1}) = 0.816$ (rather high).

Evaluation of quality of candidate distribution $\tilde{\theta} \sim N(\theta_{i-1}, 0.01^2)$ with smaller candidate steps:

- **'trace plot'** of (accepted) draws $\theta_0, \theta_1, \theta_2, \dots$:



- **acceptance percentage:** what percentage of the candidate draws is accepted?
Here: acceptance percentage = 91.5% (very high).
- **(first order) serial correlation in sequence of (accepted) draws.**
Here: $\text{corr}(\alpha_{1,i}, \alpha_{1,i-1}) = 0.967$ (very high).

Note again:

- A high acceptance percentage does **not** immediately imply a good quality of (the stdev of) the candidate distribution in the random walk Metropolis(-Hastings) method:

If variance of candidate distribution $\rightarrow 0$, then

- $\tilde{\theta} \approx \theta_{i-1}$
- $P(\tilde{\theta}) \approx P(\theta_{i-1})$
- $\alpha = \min\left\{\frac{P(\tilde{\theta})}{P(\theta_{i-1})}, 1\right\} \approx 1$
- acceptance percentage $\rightarrow 100\%$.

But then also serial correlation $\rightarrow 1$.

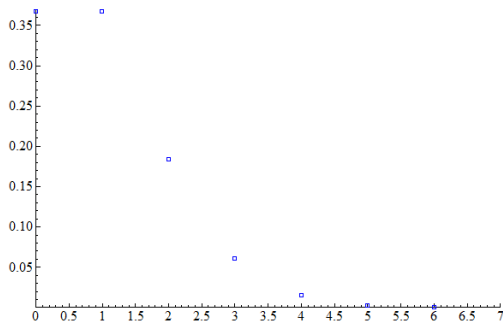
Then the random walk Metropolis(-Hastings) method is very **inefficient**: a huge number of draws may be required to 'cover' the whole posterior distribution.

Poisson distribution:

Poisson distribution with mean $E(y_j|\mu) = \mu$: probability mass function:

$$p(y_j|\mu) = \frac{\mu^{y_j} \exp(-\mu)}{y_j!} \quad y_j = 0, 1, 2, \dots$$

Example: Poisson($\mu = 1$) distribution:



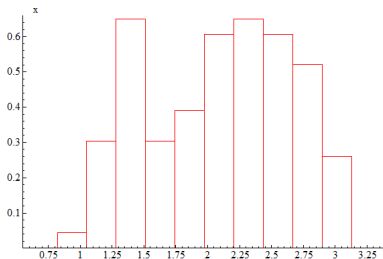
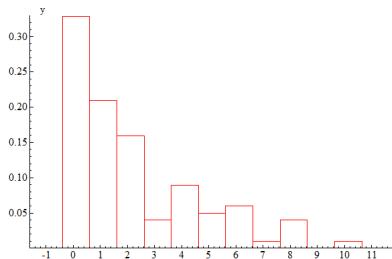
Note: for small mean μ the Poisson(μ) distribution can obviously **not** be approximated well by a normal distribution.

Poisson regression model:

$$y_j \sim \text{Poisson}(\mu_j) \quad \text{with} \quad \mu_j = \exp(\beta_0 + \beta_1 x_j).^1$$

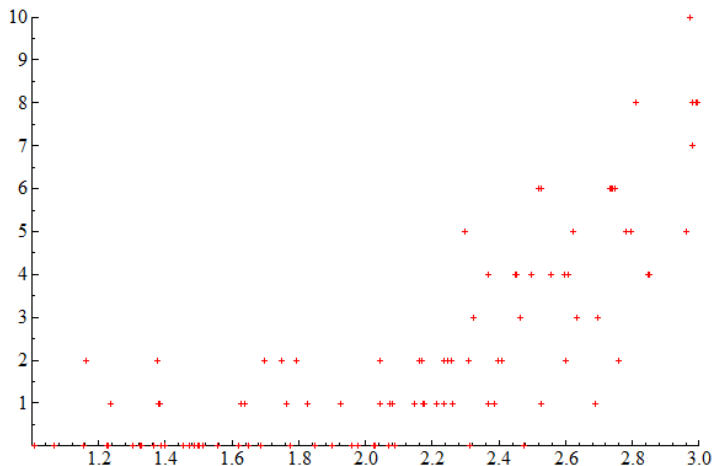
Note: exponent to ensure that $\mu_j > 0$.

For example, sales y_j (left) and advertisement expenditure x_j (right) ($j = 1, 2, \dots, n$) with $n = 100$:

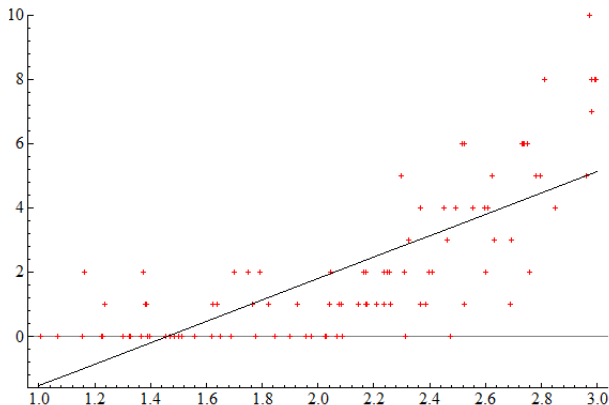


¹Compare: normal linear regression model has $y_j \sim N(\mu_j, \sigma^2)$ with $\mu_j = \beta_0 + \beta_1 x_j$. Note: everything is conditional on exogenous explanatory variable x_j ; $y_j \sim \dots$ is written instead of $y_j|x_j \sim \dots$ only to have a shorter/simpler notation here and below.

For example, sales y_j (vertical axis) and advertisement expenditure x_j (horizontal axis) ($j = 1, 2, \dots, n = 100$).



Ordinary Least Squares (OLS) may yield negative fitted values $\hat{y}$ for certain values of x :



Note: we can **not** estimate normal regression model for $\ln(y_j)$, because of observations with $y_j = 0$. And the normality assumption for $\ln(y_j)$ or $\ln(1 + y_j)$ would be unrealistic (for example, for prediction interval of y_j).

Poisson probability mass function with mean $\mu_j = \exp(\beta_0 + \beta_1 x_j)$:

$$\begin{aligned}
 p(y_j|\theta) &= \frac{\mu_j^{y_j} \exp(-\mu_j)}{y_j!} \\
 &\propto \mu_j^{y_j} \exp(-\mu_j) \\
 &= \exp(\beta_0 + \beta_1 x_j)^{y_j} \exp(-\exp(\beta_0 + \beta_1 x_j))
 \end{aligned}$$

with $\theta = (\beta_0, \beta_1)'$, where everything is conditional on exogenous $x = (x_1, \dots, x_n)'$.

Likelihood:

$$\begin{aligned}
 p(y|\theta) &= p(y_1, \dots, y_n|\theta) = \prod_{j=1}^n p(y_j|\theta) \\
 &\propto \prod_{j=1}^n \exp(\beta_0 + \beta_1 x_j)^{y_j} \exp(-\exp(\beta_0 + \beta_1 x_j))
 \end{aligned}$$

Suppose we specify non-informative prior: $p(\theta) \propto 1$ for $\theta = (\beta_0, \beta_1)'$ (with $-\infty < \beta_0 < \infty$, $-\infty < \beta_1 < \infty$).

Posterior:

$$p(\theta|y) \propto p(\theta)p(y|\theta) \propto \prod_{j=1}^n \exp(\beta_0 + \beta_1 x_j)^{y_j} \exp(-\exp(\beta_0 + \beta_1 x_j))$$

This is **not** a well-known 'standard' distribution for (β_0, β_1) .

For analyzing the posterior we need simulation method like random walk Metropolis(-Hastings) method. Use logarithm of posterior density kernel $P(\theta)$

$$\ln P(\theta) = \sum_{j=1}^n \{y_j(\beta_0 + \beta_1 x_j) - \exp(\beta_0 + \beta_1 x_j)\}.$$

Random walk Metropolis(-Hastings) method

- Choose feasible initial value θ_0 .
(For example, $(\ln(\bar{y}), 0)$ or the ML estimator.)
- Do for draw $i = 1, \dots, n_{draws}$:
 - Simulate candidate draw $\tilde{\theta}$ from candidate density $Q(\cdot)$ with mean θ_{i-1} . For example, $\tilde{\theta} \sim N(\theta_{i-1}, \Sigma)$.
 - Compute acceptance probability

$$\alpha = \min \left\{ \frac{P(\tilde{\theta})}{P(\theta_{i-1})}, 1 \right\} = \min \left\{ \exp(\ln P(\tilde{\theta}) - \ln P(\theta_{i-1})), 1 \right\}.$$

- Simulate U from uniform distribution on $[0, 1]$.
- If $U \leq \alpha$, then accept: $\theta_i = \tilde{\theta}$ (accept candidate draw).
If $U > \alpha$, then reject: $\theta_i = \theta_{i-1}$ (repeat previous draw).

Common choice of Σ (variance-covariance matrix of candidate distribution) is $-Hessian^{-1}$, where $Hessian$ is the Hessian of the log posterior at the posterior mode, which is here (with flat prior density) the same as $-Hessian^{-1}$ of the loglikelihood at the maximum likelihood estimator:

$$\left(-\frac{\partial^2 \log p(y|\theta)}{\partial \theta \partial \theta'} \bigg|_{\theta=\hat{\theta}_{ML}} \right)^{-1}.$$

Here:

$$\Sigma = \begin{pmatrix} 0.2085 & -0.0796 \\ -0.0796 & 0.0311 \end{pmatrix}.$$

Note: classical/frequentist statistician has for large enough number of observations n (under regularity conditions):

$$\hat{\theta}_{ML} \approx N \left(\theta, \left(-\frac{\partial^2 \log p(y|\theta)}{\partial \theta \partial \theta'} \bigg|_{\theta=\hat{\theta}_{ML}} \right)^{-1} \right).$$

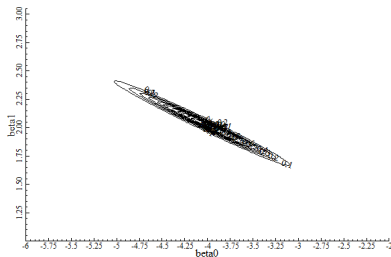
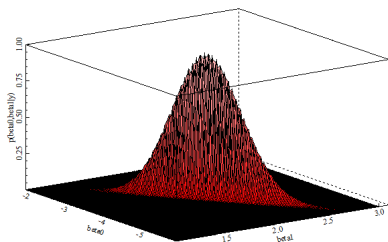
Simulation of $\tilde{\theta}$ from $N(\theta_{i-1}, \Sigma)$, the 2-dimensional normal distribution with mean θ_{i-1} and variance-covariance matrix Σ :

- Use built-in function for simulation from multivariate normal distribution, if this exists.
- In Ox/OxMetrics: compute

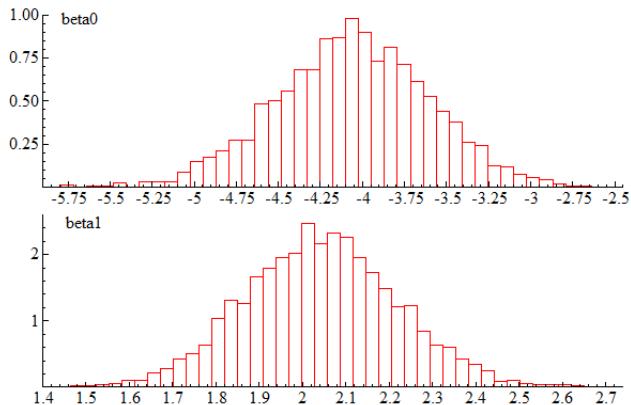
$$\tilde{\theta} = \theta_{i-1} + \text{choleski}(\Sigma)Z$$

where $Z = (Z_1, Z_2)'$ is a 2×1 vector of 2 independent standard normal $N(0,1)$ variables.

Kernel of posterior density $p(\beta_0, \beta_1 | y)$ (scaled to have maximum 1):



Histograms of 10000 draws (after burn-in of 1000 draws):



Properties of posterior distribution:

	mean	st. dev.	2.5% quantile	97.5% quantile
β_0	-4.0608	0.4577	-4.9744	-3.1738
β_1	2.0415	0.1765	1.7011	2.3919

We have:

- Acceptance percentage: 56.43%.
⇒ reasonable 'moderate' value
⇒ this suggests that there are no problems with (too small, too large, or otherwise poorly chosen) variance-covariance matrix Σ of the candidate steps.
- 95% posterior interval of β_1 is [1.7011, 2.3919] which does not include 0.
⇒ Bayesian statistician using 95% posterior interval rejects $\beta_1 = 0$.
⇒ Conclusion: There is effect of x (advertisement) on y (sales).

Purely illustrative example with uniform target density $P(\theta)$:

- Suppose we want to simulate from the uniform distribution on the interval $[0,1]$, so that the target density is

$$P(\theta) = \begin{cases} 1 & 0 \leq \theta \leq 1, \\ 0 & \text{else,} \end{cases}$$

where we use the random walk Metropolis(-Hastings) method.²

- Suppose we simulate the candidate draw $\tilde{\theta}$ from the normal distribution $N(\theta_{i-1}, \sigma_{\text{candidate}}^2)$.

²Obviously, this is only for illustrative purposes! In practice, if we want to simulate from the uniform distribution on $[0,1]$, then we **directly** simulate from this.

The acceptance probability is given by

$$\begin{aligned}\alpha &= \min \left\{ \frac{P(\tilde{\theta})}{P(\theta_{i-1})}, 1 \right\} = \\&= \min \left\{ \frac{1}{1}, 1 \right\} = 1 && \text{if } \tilde{\theta} \in [0, 1] \text{ and } \theta_{i-1} \in [0, 1] \\&= \min \left\{ \frac{0}{1}, 1 \right\} = 0 && \text{if } \tilde{\theta} \notin [0, 1] \text{ and } \theta_{i-1} \in [0, 1] \\&= \min \left\{ \frac{1}{0}, 1 \right\} = \text{undefined} && \text{if } \tilde{\theta} \in [0, 1] \text{ and } \underline{\theta_{i-1} \notin [0, 1]} \\&= \min \left\{ \frac{0}{0}, 1 \right\} = \text{undefined} && \text{if } \tilde{\theta} \notin [0, 1] \text{ and } \underline{\theta_{i-1} \notin [0, 1]}\end{aligned}$$

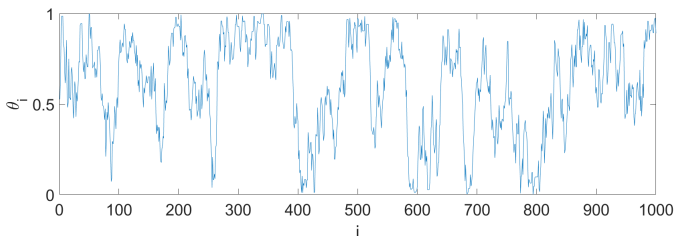
Note: the latter two cases do not occur:

- We choose initial value $\theta_0 \in [0, 1]$,
- Each candidate draw $\tilde{\theta}$ outside $[0, 1]$ is rejected.
- So, we never have $\theta_{i-1} \notin [0, 1]$.

So: we simply accept every $\tilde{\theta} \in [0, 1]$ and reject every $\tilde{\theta} \notin [0, 1]$.

How to evaluate whether a candidate distribution is 'good' or 'bad'?

- **'trace plot'** of (accepted and repeated) draws $\theta_0, \theta_1, \theta_2, \dots$:



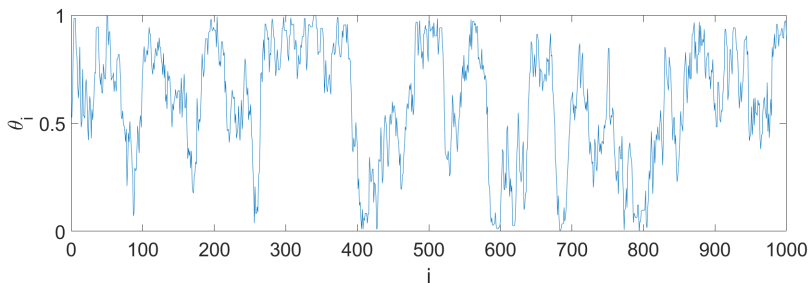
Do the draws move through the parameter space 'fast enough'?

- **acceptance percentage:** what percentage of the candidate draws is accepted? A percentage close to 0% is bad. But a percentage close to 100% can be bad too!
- **(first order) serial correlation in sequence of (accepted and repeated) draws.**
The lower the serial correlation, the better. (Close to 1 is bad.)

Case with small candidate steps: $\sigma_{candidate} = 0.1$:

$\tilde{\theta} \sim N(\theta_{i-1}, 0.1^2)$ and $n_{draws} = 100000$.

- ‘trace plot’ of first 1000 accepted (and possibly repeated) draws:

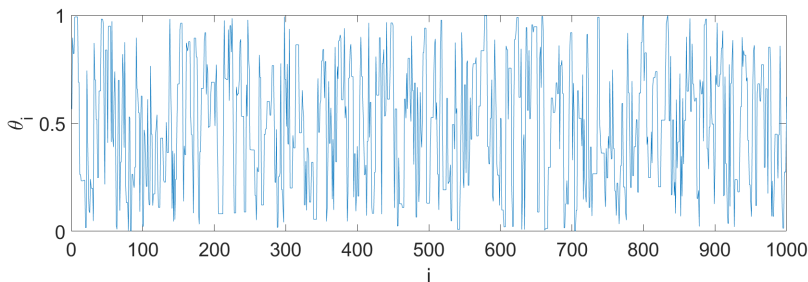


- acceptance percentage = 92%.
- first order serial correlation in sequence of accepted (and possibly repeated) draws: $\text{corr}(\theta_i, \theta_{i-1}) = 0.95$.

Case with reasonable candidate steps: $\sigma_{candidate} = 0.5$:

$\tilde{\theta} \sim N(\theta_{i-1}, 0.5^2)$ and $n_{draws} = 100000$.

- ‘trace plot’ of first 1000 accepted (and possibly repeated) draws:

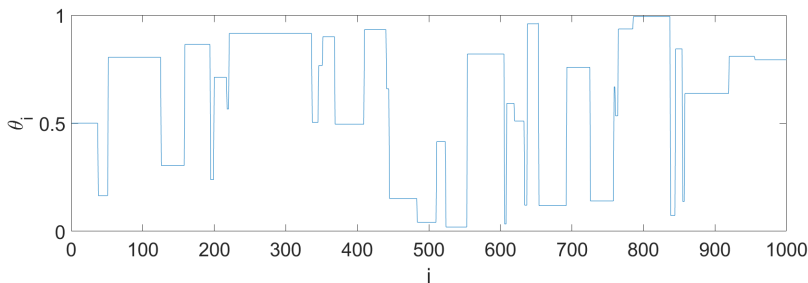


- acceptance percentage = 61%.
- first order serial correlation in sequence of accepted (and possibly repeated) draws: $\text{corr}(\theta_i, \theta_{i-1}) = 0.61$.

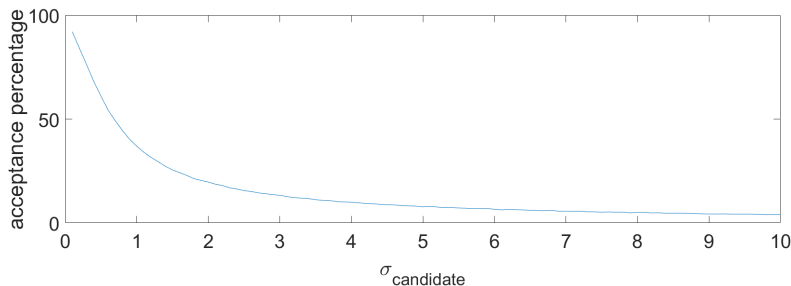
Case with large candidate steps: $\sigma_{candidate} = 10$:

$\tilde{\theta} \sim N(\theta_{i-1}, 10^2)$ and $n_{draws} = 100000$.

- ‘trace plot’ of first 1000 accepted (and possibly repeated) draws:



- acceptance percentage = 4%.
- first order serial correlation in sequence of accepted (and possibly repeated) draws: $\text{corr}(\theta_i, \theta_{i-1}) = 0.96$.



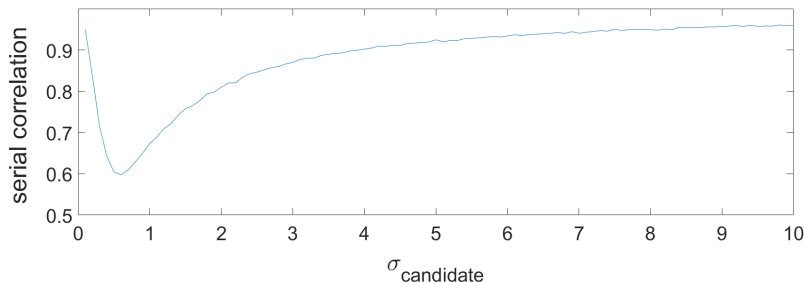
Note: For random walk Metropolis(-Hastings) method we observe that:

- very small candidate steps are often accepted.
- very large candidate steps are often rejected.

If $\sigma_{\text{candidate}} \rightarrow 0$, then $\tilde{\theta} \approx \theta_{i-1}$, $P(\tilde{\theta}) \approx P(\theta_{i-1})$, so

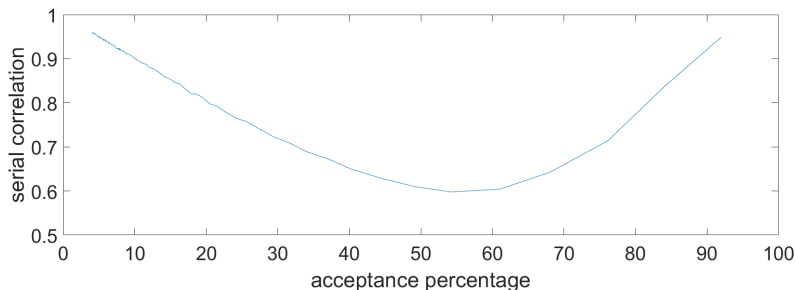
$$\alpha = \min\left\{\frac{P(\tilde{\theta})}{P(\theta_{i-1})}, 1\right\} \approx 1.$$

Then acceptance percentage $\rightarrow 100\%$, but also serial correlation $\rightarrow 1$.



Note: for random walk Metropolis(-Hastings) method we have poor performance (i.e., large serial correlation $\text{corr}(\theta_i, \theta_{i-1})$), if we have

- too small candidate steps (that move too slowly through the parameter space)
- too large candidate steps (that are mostly rejected).



Note: For random walk Metropolis(-Hastings) method **in this example**:

- Best performance (i.e., lowest serial correlation $\text{corr}(\theta_i, \theta_{i-1})$) if acceptance percentage has 'moderate' value around 50%-60%.
- Reasonable performance (reasonable serial correlation $\text{corr}(\theta_i, \theta_{i-1})$) if acceptance percentage has value between 20% and 80%.

Literature: For normal target density 23.4% is 'optimal' acceptance rate.